

DBMS – PRODUCT AND CATEGORY MANAGEMENT

Project Title: IKEA E-Commerce Database Management System

1. DATABASE DESIGN:

1.1 CATEGORY TABLE

Attribute	Description
Category_ID	Unique category ID
Category_Name	Category name
Description	Category description

1.2 PRODUCT TABLE

Attribute	Description
Product_ID	Unique product ID
Product_Name	Product name
Description	Product description
Price	Product price
Brand	Product brand
Weight	Product weight
Color	Product color
Material	Product material

Attribute	Description
Category_ID	Foreign key referencing Category table
Supplier_ID	Unique supplier ID

2. PRIMARY AND FOREIGN KEY RELATIONSHIP

Foreign Key (Child Table)	Foreign Table (Parent Table)	Referenced Column	Relationship / Constraint Description
Product.Category_ID	Category	Category_ID	Many-to-One: Links each product to a specific category. Ensures that a product cannot be assigned to a Category_ID that does not exist in the Category table.
Product.Supplier_ID	Supplier	Supplier_ID	Many-to-One: Links each product to a supplier. Ensures that Supplier_ID references a valid supplier record in the parent Supplier table.

3. TABLE CREATION

3.1 CREATION OF CATEGORY TABLE

```
CREATE TABLE Category (  
    Category_ID INT PRIMARY KEY,  
    Category_Name VARCHAR2(100),  
    Description VARCHAR2(255)  
);
```

3.2 CREATION OF PRODUCT TABLE

```
CREATE TABLE Product (  
    Product_ID INT PRIMARY KEY,  
    Product_Name VARCHAR2(100),  
    Description VARCHAR2(255),  
    Price NUMBER(10,2),  
    Brand VARCHAR2(100),  
    Weight NUMBER(8,2),  
    Color VARCHAR2(50),  
    Material VARCHAR2(100),  
    Category_ID INT,  
    Supplier_ID INT  
);
```

4. DATA INSERTION

4.1 PRODUCT RECORDS

```
INSERT INTO Product
(Product_ID, Product_Name, Description, Price, Brand, Weight, Color, Material, Category_ID, Supplier_ID)
VALUES
(181, 'KALLAX Shelf Unit', 'Versatile shelf unit for storage and display', 4999.00, 'IKEA', 18.50, 'White', 'Particleboard', 7, 1);

INSERT INTO Product
(Product_ID, Product_Name, Description, Price, Brand, Weight, Color, Material, Category_ID, Supplier_ID)
VALUES
(182, 'MALM Bed Frame', 'Simple modern bed frame for bedrooms', 12999.00, 'IKEA', 35.00, 'White', 'Wood', 2, 2);

INSERT INTO Product
(Product_ID, Product_Name, Description, Price, Brand, Weight, Color, Material, Category_ID, Supplier_ID)
VALUES
(183, 'LACK Coffee Table', 'Minimal coffee table for living rooms', 2499.00, 'IKEA', 8.30, 'Black', 'Particleboard', 1, 3);

INSERT INTO Product
(Product_ID, Product_Name, Description, Price, Brand, Weight, Color, Material, Category_ID, Supplier_ID)
VALUES
(184, 'POANG Armchair', 'Comfortable armchair with curved design', 6999.00, 'IKEA', 12.50, 'Brown', 'Wood', 1, 4);

INSERT INTO Product
(Product_ID, Product_Name, Description, Price, Brand, Weight, Color, Material, Category_ID, Supplier_ID)
VALUES
(185, 'MICKE Desk', 'Compact desk suitable for home offices', 5999.00, 'IKEA', 21.00, 'White', 'Particleboard', 5, 5);

INSERT INTO Product
(Product_ID, Product_Name, Description, Price, Brand, Weight, Color, Material, Category_ID, Supplier_ID)
VALUES
(186, 'ENCKEL Dining Chair', 'Classic dining chair with comfortable seating', 3999.00, 'IKEA', 7.80, 'Black', 'Wood', 4, 6);

INSERT INTO Product
(Product_ID, Product_Name, Description, Price, Brand, Weight, Color, Material, Category_ID, Supplier_ID)
VALUES
(187, 'FADO Table Lamp', 'Decorative table lamp with soft lighting', 1499.00, 'IKEA', 1.30, 'White', 'Glass', 8, 7);

INSERT INTO Product
(Product_ID, Product_Name, Description, Price, Brand, Weight, Color, Material, Category_ID, Supplier_ID)
VALUES
(188, 'MULIG Clothes Rack', 'Simple rack for organizing clothes', 1999.00, 'IKEA', 3.50, 'White', 'Steel', 7, 8);

INSERT INTO Product
(Product_ID, Product_Name, Description, Price, Brand, Weight, Color, Material, Category_ID, Supplier_ID)
VALUES
(189, 'ÄPPLARÖ Outdoor Table', 'Outdoor table suitable for balconies and patios', 8999.00, 'IKEA', 24.00, 'Brown', 'Acacia Wood', 6, 9);

INSERT INTO Product
(Product_ID, Product_Name, Description, Price, Brand, Weight, Color, Material, Category_ID, Supplier_ID)
VALUES
(190, 'FEMKA Artificial Plant', 'Low-maintenance artificial decorative plant', 799.00, 'IKEA', 0.80, 'Green', 'Plastic', 10, 10);

COMMIT;
```

4.2 CATEGORY RECORDS

```
INSERT INTO Category (Category_ID, Category_Name, Description)
VALUES (1, 'Living Room', 'Furniture and accessories for living rooms');

INSERT INTO Category (Category_ID, Category_Name, Description)
VALUES (2, 'Bedroom', 'Furniture and accessories for bedrooms');

INSERT INTO Category (Category_ID, Category_Name, Description)
VALUES (3, 'Kitchen', 'Kitchen furniture and storage products');

INSERT INTO Category (Category_ID, Category_Name, Description)
VALUES (4, 'Dining', 'Dining tables, chairs and accessories');

INSERT INTO Category (Category_ID, Category_Name, Description)
VALUES (5, 'Office', 'Furniture and accessories for workspaces');

INSERT INTO Category (Category_ID, Category_Name, Description)
VALUES (6, 'Outdoor', 'Furniture and products for outdoor spaces');

INSERT INTO Category (Category_ID, Category_Name, Description)
VALUES (7, 'Storage', 'Storage solutions and organizers');

INSERT INTO Category (Category_ID, Category_Name, Description)
VALUES (8, 'Lighting', 'Lamps and lighting products');

INSERT INTO Category (Category_ID, Category_Name, Description)
VALUES (9, 'Bathroom', 'Bathroom furniture and accessories');

INSERT INTO Category (Category_ID, Category_Name, Description)
VALUES (10, 'Home Decor', 'Decorative products and accessories');

COMMIT;
```

4.3 UPDATE AND DELETE FOR PRODUCT

```
Commit complete.

UPDATE Category
SET Description = 'Ergonomic furniture and accessories for modern home offices'
WHERE Category_ID = 5;

DELETE FROM Category
WHERE Category_ID = 9;
```

4.4 UPDATE FOR PRODUCT

```
UPDATE Product
SET Price = 4499.00,
    Description = 'Updated: Versatile shelf unit for storage and display'
WHERE Product_ID = 101;
```

4.5 DELETE FOR PRODUCT

```
DELETE FROM Product
WHERE Material = 'Plastic'
    AND Price < 1000.00;
```

PRODUCT_NAME

DESCRIPTION

PRICE

BRAND

WEIGHT COLOR

MATERIAL

CATEGORY_ID SUPPLIER_ID

6

9

PRODUCT_ID

PRODUCT_NAME

DESCRIPTION

PRICE

BRAND

WEIGHT COLOR

MATERIAL

CATEGORY_ID SUPPLIER_ID

9 rows selected.